

MME8215: Operations Research - Seminar

VI. Problems.

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1 Linear Programming

1.1 Renting Warehouses

Problem definition A firm has realised that it will not have sufficient storage space for the next three months. Details of the additional storage requirements for the next 3 months are provided in Table 1.

Table 1: Additional storage space requirements.

Month	January	February	March
Space required (1,000 m ²)	25	10	20

To cover its requirements, the firm plans to rent additional space on a short-term basis. At the beginning of each month, the firm can rent any amount of space for any number of months. It can pay separate rents of different quantities of space and/or over different periods of time. For instance, during the first month it could rent 20000 m² for 2 months, and rent 5000 m² separately

for 1 month. It could also acquire new rented spaces before the previous ones expire. The costs per 1000 m^2 of rented space in accordance with duration are provided in Table 2

Table 2: Rent costs.

Rent duration	1 month	2 months	3 months
Cost (\$ per 1,000 m^2)	280	450	600

Build a linear programming model whose solution provides a rent policy that covers space requirements at a minimum cost.

Solution

Decision variables:

- X_{11} = Quantity of space (m^2) the firm rents in month 1 for month 1.
- X_{12} = Quantity of space (m^2) the firm rents in month 1 for month 2.
- X_{13} = Quantity of space (m^2) the firm rents in month 1 for month 3.
- X_{21} = Quantity of space (m^2) the firm rents in month 2 for month 1.
- X_{22} = Quantity of space (m^2) the firm rents in month 2 for month 2.
- X_{31} = Quantity of space (m^2) the firm rents in month 3 for month 1.

Objective function:

$$\text{To minimise } z = 280(X_{11} + X_{21} + X_{31}) + 450(X_{12} + X_{22}) + 600X_{13}$$

Constraints:

$$\begin{aligned} X_{11} + X_{12} + X_{13} &\geq 25000 \\ X_{12} + X_{13} + X_{21} + X_{22} &\geq 10000 \\ X_{13} + X_{22} + X_{31} &\geq 20000 \\ X_{11}, X_{12}, X_{13}, X_{21}, X_{22}, X_{31} &\geq 0 \end{aligned}$$

The objective function minimises the cost of the space to be rented. The first three constraints ensure that the quantity of space required is rented for months 1, 2 and 3, respectively. The last constraint imposes non-negativity to all the variables.

1.2 Crop Planning

Problem definition A citrus firm in Valencia possesses 120 acres (1 acre = 4074 m^2) and it is planning to sow at least three crops. The seeds for crops A, B and C cost \$40, \$20 and \$30 per acre, respectively. It intends to invest a maximum of \$3400 in seeds. Seeds A, B and C require 1, 2 and 1 working day(s) per acre, respectively, and 170 working days available are considered. If the owner can obtain a profit of \$100 per acre with seed A, \$300 per acre with seed B and \$200 per acre with seed C, how many acres should the owner sow of each seed to maximise the profit?

1. Model the programme like a linear programming model.
2. Is it possible to sow 40 acres with seed A, 50 acres with seed B and 30 acres with seed C?
3. Is it possible to not sow seed A, and to sow 40 acres with seed B and 70 acres with seed C?
4. What is the optimum solution? What is the optimum objective function value?
5. What capacity is completely used with the optimum combination?
6. Where are the idle capacities?

7. If the budget was increased by \$1, how much would the total profit increase by? Why?
8. If the working days available increased by one extra day, how much would the total profit increase by? Why?
9. If the initial budget was changed to \$3401, what would happen to the optimum objective function value?
10. If the budget was limited to \$800 for seed B, what would happen to the optimum objective function value?
11. If the profit per acre with seed B was \$400, would the optimum amounts in acres be the same? Why?
12. If the profit per acre with seed C were \$310, would the optimum amounts in acres be the same? Why?

Solution

Table 3: Solution obtained for problem 2.

Decision variable	Solution value	Unit cost or profit c(j)	Total contribution	Reduced cost	Basic status	Allowable Min. c(j)	Allowable Max. c(j)
X_1	0	100.0000	0	-100.0000	At bound	-M	200.0000
X_2	50.0000	300.0000	15000.0000	0	Basic	200.0000	400.0000
X_3	70.0000	200.0000	14000.0000	0	Basic	150.0000	300.0000
Objective Function	Max=	29000.0000					
Constraint	Left-hand Side	Direction	Right-hand side	Slack or Surplus	Shadow price	Allowable Min. RHS.	Allowable Max. RHS.
C1	120.0000	$\leq$	120.0000	0	100.0000	85.0000	127.5000
C2	3100.0000	$\leq$	3400.0000	300.0000	0	3100.0000	M
C3	170.0000	$\leq$	170.0000	0	100.0000	140.0000	240.0000

1. Decision variables:

- X_1 = number of acres to sow with seed A.
- X_2 = number of acres to sow with seed B.
- X_3 = number of acres to sow with seed C.

Objective function (maximise the profit):

$$\text{Max } z = 100X_1 + 300X_2 + 200X_3$$

Constraints:

$$\begin{aligned}
 X_1 + X_2 + X_3 &\leq 120 && \text{(total number of acres available)} \\
 40X_1 + 20X_2 + 30X_3 &\leq 3400 && \text{(budget available for seeds)} \\
 X_1 + 2X_2 + X_3 &\leq 170 && \text{(number of working days available)} \\
 X_1 + X_2 + X_3 &\geq 0
 \end{aligned}$$

The solution to the proposed problem has been obtained with a commercial solver (see the sample in Table 3).

2. It would not be the optimum solution.

3. It would be the optimum solution.
4. Not sowing seed A, sowing 40 acres with seed B and 70 acres with seed C to obtain a profit of \$29000.
5. The total number of acres available and the total number of working days.
6. In the available budget for seeds.
7. It would increase by nothing because it is an idle capacity.
8. By \$100, which corresponds to the shadow price of the corresponding constraint.
9. Nothing because there is already a surplus of \$300.
10. It would be necessary to add a new constraint to the initial model, $20X_2 \leq 800$, and the problem would have to be solved again in order to answer this question.
11. All we can state with any certainty is that the optimum solution structure would be the same. This is because 400 is within the limit of the ranges for the profit coefficient of X_2 .
12. All we can state with any certainty is that the optimum solution structure would be the same. This is because 310 is within the limit of the ranges for the profit coefficient of X_3 .

2 Integer programming

2.1 Distribution of Air-Conditioning Units

Problem definition A North American air-conditioning firm has production plants in Portland and Flint. It has to supply a certain number of units in its distribution centres located in Los Angeles and Atlanta. The delivery costs are summarised in Table 4. The supply and demand data

Table 4: Delivery costs between distribution centres.

Production plants	Distribution centres	Delivery costs
Portland	Los Angeles	\$30
	Atlanta	\$40
Flint	Los Angeles	\$60
	Atlanta	\$50

are provided in Table 5 as number of units:

Table 5: Supply and demand data.

Supply	Available units	Demand	Required units
Portland	200	Los Angeles	300
Flint	600	Atlanta	400

1. Consider an integer linear programming model that determines how the deliveries from Portland and Flint should be carried out in order to minimise delivery costs.
2. Solve the problem and interpret the results.

Solution

a. Decision variables:

- X_1 = Units delivered from Portland to Los Angeles.

- X_2 = Units delivered from Portland to Atlanta.
- X_3 = Units delivered from Flint to Los Angeles.
- X_4 = Units delivered from Flint to Atlanta.

Objective function:

$$\text{Minimize } z = 30X_1 + 40X_2 + 60X_3 + 50X_4$$

Constraints:

$$X_1 + X_2 \leq 200$$

$$X_3 + X_4 \leq 600$$

$$X_1 + X_3 \geq 300$$

$$X_2 + X_4 \geq 400$$

$$X_1, X_2, X_3, X_4 \geq 0 \text{ and integer values}$$

b. Step 1: Standardising the problem

A minimisation problem is dealt with by multiplying the objective function by -1 and we transform it into a maximisation problem.

$$\text{Min } z = \text{Max } w = -30X_1 - 40X_2 - 60X_3 - 50X_4 + 0S_1 + 0S_2 - 0S_3 - 0S_4 - MA_3 - MA_4$$

$$X_1 + X_2 + S_1 = 200$$

$$X_3 + X_4 + S_2 = 600$$

$$X_1 + X_3 - S_3 + A_3 = 300$$

$$X_2 + X_4 - S_4 + A_4 = 400$$

Step 2: Generating an initial solution (Tables 6, 7, 8 and 9).

Table 6: Simplex Method.

C_j	C_j Base	-30 X_1	-40 X_2	-60 X_3	-50 X_4	0 S_1	0 S_2	0 S_3	0 S_4	-M A_3	-M A_4	R	Ratio
0	S_1	1	1	0	0	1	0	0	0	0	0	200	200
0	S_2	0	0	1	1	0	1	0	0	0	0	600	600/0
-M	A_3	1	0	1	0	0	0	-1	0	1	0	300	300
-M	A_4	0	1	0	1	0	0	0	-1	0	1	400	400/0
	$C_j - Z_j$	-30+M	-40+M	-60+M	-50+M	0	0	-M	-M	0	0		

Table 7: Simplex Method.

C_j	C_j Base	-30 X_1	-40 X_2	-60 X_3	-50 X_4	0 S_1	0 S_2	0 S_3	0 S_4	-M A_3	-M A_4	R	Ratio
-30	X_1	1	1	0	0	1	0	0	0	0	0	200	200/0
0	S_2	0	0	1	1	0	1	0	0	0	0	600	600
-M	A_3	0	-1	1	0	-1	0	-1	0	1	0	100	100/0
-M	A_4	0	1	0	1	0	0	0	-1	0	1	400	400
	$C_j - Z_j$	0	-10	-60+M	-50+M	30-M	0	-M	-M	0	0		

$$X_1 = X_2 = X_3 = X_4 = S_3 = S_4 = 0$$

The solution to the problem is:

$$\begin{aligned} X_1 &= 200 \\ X_2 &= 0 \\ X_3 &= 100 \\ X_4 &= 400 \\ z &= 32000 \end{aligned}$$

Table 8: Simplex Method.

	C_j	-30	-40	-60	-50	0	0	0	0	-M	-M	
C_j	Base	X_1	X_2	X_3	X_4	S_1	S_2	S_3	S_4	A_3	A_4	R
-30	X_1	1	1	0	0	1	0	0	0	0	0	200
0	S_2	0	-1	1	0	0	1	0	1	0	-1	200
-M	A_3	0	-1	1	0	-1	0	-1	0	1	0	100
-50	X_4	0	1	0	1	0	0	0	-1	0	1	400
	$C_j - Z_j$	0	40 - M	-60 + M	0	30 - M	0	-M	-50	0	-M + 50	

Table 9: Simplex Method.

	C_j	-30	-40	-60	-50	0	0	0	0	-M	-M	
C_j	Base	X_1	X_2	X_3	X_4	S_1	S_2	S_3	S_4	A_3	A_4	R
-30	X_1	1	1	0	0	1	0	0	0	0	0	200
0	S_2	0	0	0	0	1	1	1	1	-1	-1	200
-60	X_3	0	-1	1	0	-1	0	-1	0	1	0	100
-50	X_4	0	1	0	1	0	0	0	-1	0	1	400
	$C_j - Z_j$	0	-20	0	0	-30	0	-60	-50	60 - M	50 - M	

In other words, send 200 units from Portland to Los Angeles, 100 go from Flint to Los Angeles, and 400 leave Flint to Atlanta, with a total cost of \$32,000.

2.2 Allocation in a Lawyers' Office

Problem definition A lawyers' office has taken on five new cases, and each one can be adequately handled by any of the five more recent partners. Given the difference in experience and practice, lawyers will spend different times on their cases. One of the most experienced partners has estimated requirements in terms of time (in hours; see Table [11](#)).

Table 10: Estimation of the time required per case and lawyer.

	Case 1	Case 2	Case 3	Case 4	Case 5
Lawyer 1	45	22	30	65	15
Lawyer 2	180	163	85	148	68
Lawyer 3	21	17	193	59	65
Lawyer 4	18	83	16	70	115
Lawyer 5	97	75	20	70	71

In order to determine the optimum way of allocating cases to lawyers so that each one takes on a different case and that the total time of hours spent is the minimum formulate this problem as an integer linear programming model.

Solution

Decision variables:

This problem can be modelled as an allocation problem. As it is a matter of deciding which lawyer takes on each case, we define X_{ij} which takes a value of 1 if lawyer i is allocated case j , and 0 otherwise, where $i, j = 1, \dots, 5$. These 25 variables are the decision variables of the problem.

Objective function:

Our objective is to optimally allocate lawyers to cases (or vice versa). Evidently, one allocation is better than another if the total number of hours spent on preparing cases is lower. Hence, the objective function must minimise the number of hours that the five lawyers spend on preparing the five cases.

$$\begin{aligned} \text{Minimize } z = & 45X_{11} + 22X_{12} + 30X_{13} + 65X_{14} + 15X_{15} + 180X_{21} + 163X_{22} \\ & + 85X_{23} + 148X_{24} + 68X_{25} + 21X_{31} + 17X_{32} + 193X_{33} + 59X_{34} \\ & + 65X_{35} + 18X_{41} + 83X_{42} + 16X_{43} + 70X_{44} + 115X_{45} + 97X_{51} \\ & + 75X_{52} + 20X_{53} + 70X_{54} + 71X_{55} \end{aligned}$$

Constraints:

The constraints of this problem emerge because each lawyer is expected to take on a different case. If lawyer 1 can take on only one case, then one, and only one, of variables $X_{1j}, j = 1, \dots, 5$ takes a value of 1, while the other four should take a value of zero:

$$\begin{aligned} X_{11} + X_{12} + X_{13} + X_{14} + X_{15} &= 1 \text{ (1 case for lawyer 1)} \\ X_{21} + X_{22} + X_{23} + X_{24} + X_{25} &= 1 \text{ (1 case for lawyer 2)} \\ X_{31} + X_{32} + X_{33} + X_{34} + X_{35} &= 1 \text{ (1 case for lawyer 3)} \\ X_{41} + X_{42} + X_{43} + X_{44} + X_{45} &= 1 \text{ (1 case for lawyer 4)} \\ X_{51} + X_{52} + X_{53} + X_{54} + X_{55} &= 1 \text{ (1 case for lawyer 5)} \end{aligned}$$

Besides, together with the fact that there are as many lawyers as cases, each case can be prepared by only one lawyer:

$$\begin{aligned} X_{11} + X_{21} + X_{31} + X_{41} + X_{51} &= 1 \text{ (1 lawyer for case 1)} \\ X_{12} + X_{22} + X_{32} + X_{42} + X_{52} &= 1 \text{ (1 lawyer for case 2)} \\ X_{13} + X_{23} + X_{33} + X_{43} + X_{53} &= 1 \text{ (1 lawyer for case 3)} \\ X_{14} + X_{24} + X_{34} + X_{44} + X_{54} &= 1 \text{ (1 lawyer for case 4)} \\ X_{15} + X_{25} + X_{35} + X_{45} + X_{55} &= 1 \text{ (1 lawyer for case 5)} \\ X_{ij} &\in \{0, 1\}, i, j = 1 \dots 5 \end{aligned}$$

3 Non-Linear programming

3.1 Placing a Handrail

Problem definition The intention is to place a handrail on a terrace with an empty space in its centre in such a way that the handrail goes around the interior and exterior terrace perimeters. This empty space, which is rectangular in shape, measures 10 m wide by 18 m long (see Figure 1). Between the interior and the exterior terrace perimeters, a walkway must be left, which is the same width on all sides. If the handrail measures 250m:

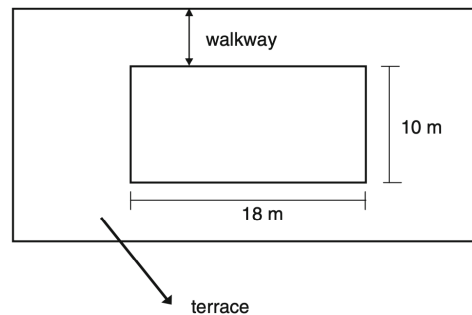


Figure 1: Sketch of the terrace.

1. Formulate a non-linear programming model that maximises the area occupied by the walkway to improve circulation in it.
2. Apply the Kuhn-Tucker conditions to the model in the previous section and indicate the metres that the outer terrace sides and the walkway width measure.
3. This problem could have been formulated by linear programming and by maximising the walkway width in this case, but how?

Solution

a. Decision variables:

-
- X_1 metres that the outer terrace side measures in parallel with the 10-m side of the empty space.
- X_2 metres that the outer terrace side measures in parallel with the 18-m side of the empty space.

Objective function (maximize the walkway area):

$$\text{Max } f = X_1 X_2 - 180 \text{ rectangular outer area} - \text{empty space}$$

Constraints:

$$\begin{aligned} (X_1 - 10)/2 &= (X_2 - 18)/2 && \text{(The walkway width is equal on all sides)} \\ 2(X_1 - 10)(X_2 - 18) &\leq 250 && \text{(Only a 250-m handrail is available)} \\ X_1, X_2 &\geq 0 \end{aligned}$$

b. The canonical form of the problem:

$$\begin{aligned} \text{Max } f &= X_1 X_2 - 180 \\ 0.5X_1 - 0.5X_2 + 4 &\leq 0 \\ 0.5X_1 + 0.5X_2 - 4 &\leq 0 \\ 2X_1 + 2X_2 - 194 &\leq 0 \end{aligned}$$

The Lagrangian function:

$$\begin{aligned} L &= X_1 X_2 - 180 + \lambda_1(0.5X_1 - 0.5X_2 + 4) + \lambda_2(-0.5X_1 + 0.5X_2 - 4) \\ &\quad + \lambda_3(2X_1 + 2X_2 - 194) \end{aligned}$$

Kuhn-Tucker conditions:

Gradient condition:

$$\begin{aligned} \frac{\partial L}{\partial X_1} &= X_2 + 0.5\lambda_1 - 0.5\lambda_2 + 2\lambda_3 = 0 \\ \frac{\partial L}{\partial X_2} &= X_1 - 0.5\lambda_1 + 0.5\lambda_2 + 2\lambda_3 = 0 \end{aligned}$$

Feasibility condition:

$$\begin{aligned} 5X_1 - 0.5X_2 + 4 &\leq 0 \\ -0.5X_1 + 0.5X_2 - 4 &\leq 0 \\ 2X_1 + 2X_2 - 194 &\leq 0 \end{aligned}$$

Orthogonality condition:

$$\begin{aligned} \lambda_1(0.5X_1 - 0.5X_2 + 4) &= 0 \\ \lambda_2(-0.5X_1 + 0.5X_2 - 4) &= 0 \\ \lambda_3(2X_1 + 2X_2 - 194) &= 0 \end{aligned}$$

Non-positivity condition

$$\lambda_1, \lambda_2, \lambda_3 \leq 0$$

Solution:

If $\lambda_1 = \lambda_2 = \lambda_3 = 0 \rightarrow X_1 = X_2 = 0$ (not possible)
 If $\lambda_1 = \lambda_2 = 0$ and $\lambda_3 \neq 0 \rightarrow X_1 = X_2$ (not possible)
 If $\lambda_1 = \lambda_3 = 0$ and $\lambda_2 \neq 0 \rightarrow X_1 = -X_2$ (not possible)
 If $\lambda_1 = \lambda_2 \neq 0$ and $\lambda_3 = 0 \rightarrow X_1 = -X_2$ (not possible)
 If $\lambda_1 = \lambda_2 = \lambda_3 \neq 0$:
 $(0.5X_1 - 0.5X_2 + 4) = 0$
 $(-0.5X_1 + 0.5X_2 - 4) = 0$
 $(2X_1 + 2X_2 - 194) = 0.$

There are three equations with two unknown quantities, where $X_1 = 44.5$, $X_2 = 52.5$ and the walkway width is 17.25 m.

The Hessian Matrix of $f(x_1, x_2) = (0, 1, 1, 0)$. Thus f is semi-defined negative, the function is concave and, therefore, the values obtained are a global maximum of f .

c. Decision variables:

- X metres of walkway width.
- X_1 metres that the outer terrace side measures in parallel with the 10-m side of the empty space.
- X_2 metres that the outer terrace side measures in parallel with the 18-m side of the empty space.

Objective function:

$$\text{Max } f = X$$

Constraints:

$$X = (X_1 - 10)/2 \text{ (walkway width equal on all sides)}$$

$$X = (X_2 - 18)/2 \text{ (walkway width equal on all sides)}$$

$$2[(2X + 18) + (2X + 10)] + 56 \leq 250 \text{ (outer perimeter + inner perimeter smaller than or equal to 250 m)}$$

$$X, X_1, X_2 \geq 0.$$

3.2 Designing and Planning a Project to Launch a Scientific Balloon

Problem definition A research centre has to design and launch a balloon with an X-ray telescope and other scientific equipment. An approximate measure of performance can be expressed in terms of the height that the balloon reaches and the weight of the equipment raised. Evidently, height itself is given by the volume of the balloon. Based on past experience, it was concluded that to maximise a satisfactory function for performance, then:

$$P = f(V, W) = 100V - 0.3V^2 + 80W - 0.2W^2$$

where V is the volume of the balloon and W is the weight of the equipment. The project budget is limited to \$1040. The cost associated with volume V is $2V$, and the cost of the equipment is $4W$. To obtain a reasonable balance between performance owing to the height and weight of the scientific equipment, the design must fulfil constraint $80W \geq 100V$.

Obtain an optimum design in terms of the volume and weight of the equipment, and solve it using the Kuhn-Tucker algorithm.

Solution

Decision Variables:

V = volume of the balloon;
 W = weight of the equipment;

Objective Function:

$$\text{Maximise } = f(V, W) = 100V - 0.3V^2 + 80W - 0.2W^2$$

Constraints:

$$2V + 4W \leq 1040 \rightarrow 2V + 4W - 1040 \leq 0$$

$$80W \geq 100V \rightarrow 100V - 80W \leq 0$$

Non-negativity:

$$V, W \geq 0$$

Lagrangian Function:

$$L(V, W, \lambda_1, \lambda_2) = 100V - 0.3V^2 + 80W - 0.2W^2 + \lambda_1(2V + 4W - 1040) + \lambda_2(100V - 80W)$$

Gradient condition:

$$\partial L / \partial V = 100 - 0.6V + 2\lambda_1 + 100\lambda_2 = 0$$

$$\partial L / \partial W = 80 - 0.4W + 4\lambda_1 - 80\lambda_2 = 0$$

Orthogonality condition:

$$\lambda_1(2V + 4W - 1040) = 0 \quad \lambda_2(100V - 80W) = 0$$

Feasibility condition:

$$\partial L / \partial \lambda_1 = 2V + 4W - 1040 \leq 0$$

$$\partial L / \partial \lambda_2 = 100V - 80W \leq 0$$

Non-positivity condition:

$$\lambda_1, \lambda_2 \leq 0$$

By analysing the possible values of λ_1 and λ_2 , we obtain:

$$V = 148.57;$$

$$W = 185.71;$$

$$\lambda_1 = -2.57;$$

$$\lambda_2 = -0.06;$$

Based on the Hessian Matrix of f , $H = [-0.6, 0, 0, -0.4]$, it is possible to conclude that the objective function is concave. Therefore the critical point obtained is a global maximum.

4 Network Modelling

4.1 Assigning Cases in a Lawyer's Office

Problem definition A lawyer's office has accepted five new cases, and each can be suitably managed by any of the five more recent partners. Owing to the difference in experience and practice, lawyers will employ different times on their cases. One of the most experienced partners has estimated time requirements (in hours) which are indicated in Table 11.

Table 11: Estimation of the time requirements per lawyer and case.

	Case 1	Case 2	Case 3	Case 4	Case 5
Lawyer 1	45	22	30	65	15
Lawyer 2	180	163	85	148	68
Lawyer 3	21	17	193	59	65
Lawyer 4	18	83	16	70	115
Lawyer 5	97	75	20	70	71

Formulate this problem as a Graph Theory model to optimally determine the assignment of cases to lawyers so that each one is dedicated to a different case and that the total time spent in hours is minimum.

Solution

This is a maximal flow model at a minimum cost. The structure of the graph is indicated in Figure 2.

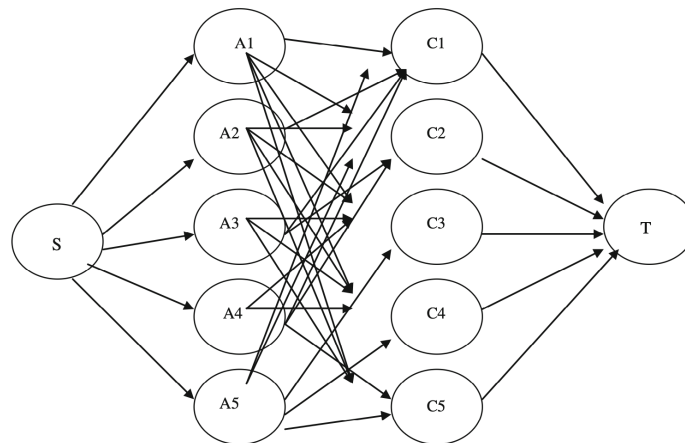


Figure 2: The assigning cases to lawyers graph.

Each arc must contain the following information:

- A_i = minimum capacity.
- B_i = maximum capacity.
- C_i = minimum cost.

In Figure 2, the arcs going from source node S to each lawyer have a minimum capacity of 0, a maximum capacity of 1 and a minimum cost of 0; that is: $(0, 1, 0)$.

The arcs going from each lawyer to each case will have a minimum capacity of 0, a maximum capacity of 1 and the cost of the corresponding hours. For example, the arc that goes from lawyer 1 to case 1, will be: $(0, 1, 145)$.

Finally, the arcs going from each case to destination node T will have a minimum capacity of 0, a maximum capacity of 1 and a minimum cost of 0; that is: $(0, 1, 0)$.